

Java Frameworks

- Spring Framework
- Hibernate
- Jakarta EE (formerly Java EE)
- Apache Struts
- JSF (JavaServer Faces)
- Dropwizard
- Micronaut
- Quarkus
- Play Framework

Spring Framework

- Enterprise applications, backend services
- Lightweight and modular
- Dependency Injection (IoC)
- Spring Boot for rapid development
- Extensive ecosystem (Spring MVC, Spring Data, Spring Security, etc.)
- REST APIs, microservices, large-scale enterprise apps

Hibernate

- ORM (Object-Relational Mapping) tool
- Maps Java objects to database tables
- Reduces boilerplate JDBC code
- Supports lazy loading and caching
- Data persistence in database-driven applications

Jakarta EE (formerly Java EE)

- Full-stack enterprise applications
- Specifications for web, transactional, and distributed systems
- Includes Servlets, JSP, EJB, JPA, etc.
- Large enterprise systems following official Java standards

Apache Struts

- Web application development (MVC)
- Based on MVC architecture
- Easy integration with other Java technologies
- Enterprise-level web applications

JSF (JavaServer Faces)

- Web interfaces (MVC-based)
- Component-based UI framework
- Managed by Jakarta EE
- Web apps with rich UI, integrated with Java backend

Dropwizard

- Microservices and REST APIs
- Combines Jetty, Jersey, Jackson, and more
- Lightweight and production-ready
- High-performance RESTful services

Micronaut

- Cloud-native and microservice apps
- Fast startup time and low memory footprint
- Dependency Injection without reflection
- AOT (Ahead-of-Time) compilation
- Serverless and reactive microservices

Quarkus

- Cloud-native Java apps, Kubernetes, and GraalVM
- Tailored for GraalVM and container deployment
- Reactive and imperative programming
- Java apps for Kubernetes, containers, and serverless environments

Play Framework

- Web apps with high productivity
- Reactive, scalable, and stateless
- Built on Akka for concurrency
- Modern web applications with Scala or Java

PHP Frameworks

- Laravel
- Symfony
- CodeIgniter
- Yii (Yes It Is!)
- Zend Framework / Laminas
- Phalcon
- Slim
- CakePHP

Laravel

- Full-featured web applications
- Elegant syntax and MVC architecture
- Built-in tools like routing, authentication, migrations, queue, testing
- Blade templating engine
- Laravel Forge, Vapor, and Eloquent ORM
- Used to develop Complex web applications, REST APIs, SaaS platforms

Symfony

- Enterprise-grade applications
- Reusable components and bundles
- Strong community and long-term support (LTS)
- Used by Drupal and Laravel internally
- Used in Large, scalable enterprise systems

CodeIgniter

- Lightweight and fast web development
- Minimal configuration
- Small footprint
- Easy to learn for beginners
- Simple and fast web apps, legacy PHP upgrades

Yii (Yes It Is!)

- High-performance web applications
- Gii code generator
- Built-in RBAC and security features
- ActiveRecord ORM
- CRUD-heavy applications, CMS, forums

Zend Framework / Laminas

- Enterprise-grade apps with high flexibility
- Now continued as Laminas Project
- Highly configurable with modular components
- Complex, high-performance apps for businesses

Phalcon

- Ultra-fast applications
- Written in C as a PHP extension
- Low overhead and high performance
- Performance-critical applications

Slim

- Microservices and REST APIs
- Lightweight micro-framework
- Focused on HTTP request/response
- RESTful APIs, microservices, lightweight apps

CakePHP

- Rapid application development
- Convention over configuration
- Built-in ORM and validation
- Prototyping and building structured apps quickly

Python Frameworks

- Django
- Flask
- FastAPI
- Pyramid

Django

- Full-stack framework
- “Batteries-included” philosophy (ORM, admin, auth, etc.)
- Highly secure and scalable
- Used for CMS, e-commerce, social media apps, admin dashboards

Flask

- Micro-framework
- Minimal and flexible
- Easy to get started, good for beginners
- REST APIs, lightweight apps, MVPs

FastAPI

- Asynchronous web framework
- High performance, automatic docs (Swagger/OpenAPI)
- Based on Python type hints
- High-performance APIs, microservices, async apps

Pyramid

- Scalable web framework
- Flexible – grow from small to large apps
- Strong security features
- Enterprise apps, CMS

Next Chapter